

PRATYAKSH TOMAR

☎ +91-8287642034 ✉ Pratyakshtomar2000@gmail.com 🔗 LinkedIn 🐙 Github 📁 Code360 📖 Leetcode 🌐 Portfolio

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, TypeScript

Web Development: FastAPI, Flask, Node.js, Bun, REST APIs, HTML5, CSS3, Tailwind CSS

AI & Machine Learning: Groq API, Pandas, Chart.js, Matplotlib, LangChain, RAG, Ollama, NLTK, Scikit-learn, TensorFlow

Databases: MongoDB, MySQL

Tools & Platforms: Git/GitHub, Vercel

Coursework: Data Structures, Algorithms, Object-Oriented Programming, Database Management System, Operating Systems, Computer Networks

PROJECTS

ClawNest | *TypeScript, Bun, OpenRouter API, Node.js*

[Live](#) | [Github](#)

- Built an open-source **personal AI agent system** in **TypeScript** on **Bun** that automates software development tasks through a terminal CLI and a remote **Telegram bot**, with **Agent**, **Plan**, and **Ask** modes.
- Engineered an **autonomous agent loop** with a modular tool executor for file operations, shell commands, code search, and web scraping, integrating **OpenRouter** to switch between multiple LLMs.
- Designed a **staged approval workflow** with git-style diff previews so every AI-proposed change is reviewed before it is applied, plus path validation, file-size limits, and owner-only Telegram access.
- Implemented an AI **planning mode** that breaks high-level goals into executable steps for selective execution, and a codebase Q&A mode that answers questions with file citations.

California Housing Predictor | *Python, Flask, scikit-learn, Chart.js, Pandas*

[Live](#) | [Github](#)

- Built a full-stack **machine learning web app** using **Flask** and **scikit-learn** that predicts California median house values with a **Random Forest Regressor** (**R^2 of 0.83**, MAE of about \$30.9K).
- Engineered a **preprocessing pipeline** (median imputation, standard scaling, one-hot encoding) and used **stratified sampling** on income categories for an 80/20 train/test split over 20,640 records.
- Developed a **REST API** with single-house prediction, **batch CSV inference** (drag-and-drop upload, summary metrics, enriched CSV export), and computed housing ratios such as rooms per household.
- Designed an interactive **evaluation dashboard** with **Chart.js** showing actual vs. predicted, residual distribution, and regional comparisons, with ocean-proximity and price filters plus worst-prediction analysis.

EDUCATION

Dr. A.P.J. Abdul Kalam Technical University, Lucknow

Bachelor of Technology in Computer Engineering

2023 – Present

Dehradun Public School, Ghaziabad

Senior Secondary (Class XII) – CBSE

2020 – 2022

Dehradun Public School, Ghaziabad

Secondary (Class X) – CBSE

2018 – 2020

ACHIEVEMENTS

- Selected as a **finalist** in **Policy Conclave '26** at **Indian Institute of Technology Kanpur**.
- Competed in the **National AI/ML Hackathon** by **AVEVA** (organised by **IIT Hyderabad**), building AI/ML solutions as part of team **Thunders**, Mar 2026.
- Participated in **Hacker's Premier League** at **Guru Gobind Singh Indraprastha University, Delhi**.
- Awarded **Top Performer certificate** in **E&ICT IITG: Introduction to Programming** by **Coding Ninjas × IIT Guwahati**, Feb 2025 (NSDC certified).
- Solved **400+** **coding problems** across platforms including **LeetCode** and **Code 360**.